

WhiteBox: Explainable Tumour Mutational Burden Prediction from Multimodal Cancer Data

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Abstract

This paper presents WhiteBox, a comprehensive machine learning framework for the continuous prediction of Tumour Mutational Burden (TMB) utilizing multimodal transcriptomic and histopathological data. Evaluated on a colon adenocarcinoma (COAD) cohort, we systematically benchmark nine distinct models across three architectural paradigms: robust transcriptomic baselines, deep multiple instance learning (MIL) networks for whole-slide images (WSI), and complex multimodal fusion architectures. To ensure computational feasibility without sacrificing methodological integrity, we introduce hardware-aware adaptations of state-of-the-art models—specifically from Shimada, Huang, and CellMorphNet—transitioning them into continuous regression tasks. Our findings demonstrate that a custom non-linear late fusion meta-learner successfully captures non-redundant spatial and genomic signals, achieving superior predictive calibration. Specifically, the late fusion framework yields an AUROC of 0.98 at the clinical ≥ 10 Mut/Mb threshold alongside a Pearson correlation of 0.73, outperforming both the standalone transcriptomic Deep Neural Network (AUROC 0.97, $r = 0.68$) and the adapted image-only networks (AUROC ≤ 0.72). Crucially, we prioritize clinical transparency by integrating a four-technique explainable AI (XAI) audit (SHAP, LIME, PDP, and Grad-CAM) tailored to this decoupled architecture. This allows for the precise visual localization of morphological drivers alongside the quantification of genomic feature reliance, directly addressing the “black-box” dilemma in medical AI. By combining state-of-the-art predictive accuracy with rigorous bias auditing and configuration-driven reproducibility, WhiteBox delivers a highly transparent, clinically aligned prototype for precision oncology.

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1 Introduction

1.1 Clinical Significance of TMB

Tumour Mutational Burden (TMB) is defined as the total number of somatic non-synonymous mutations per megabase (Mut/Mb) of coding genomic sequence. Clinically, TMB has emerged as a robust predictive biomarker for response to immune checkpoint inhibitors (ICIs) targeting PD-1/PD-L1 and CTLA-4. High-TMB tumours generate a greater diversity of neoantigens, enhancing T-cell recognition and improving progression-free survival in multiple malignancies, including non-small cell lung cancer, melanoma, and colorectal cancer [1, 3]. Despite its prognostic utility, routine TMB quantification relies on whole-exome sequencing (WES) or large targeted panels, which incur substantial costs (\$5,000–\$10,000 per sample), require specialized bioinformatics infrastructure, and introduce turnaround delays of 2–4 weeks [3]. These barriers limit equitable access to immunotherapy stratification, particularly in resource-constrained clinical settings. Consequently, there is a pressing need for cost-effective, rapid surrogate models that can infer TMB status from routinely collected clinical and histopathological data.

1.2 Motivation for Multimodal Approach

Single-modality models inherently capture only a fraction of the biological complexity driving mutational burden. Genomic and clinical tabular data encode molecular drivers (e.g., *TP53*, *KRAS* mutations), patient demographics, and disease stage, but lack spatial context. Conversely, hematoxylin and eosin (H&E) whole-slide images (WSIs) capture tissue architecture, immune infiltration, stromal composition, and morphological heterogeneity that are invisible to bulk sequencing [2]. Prior studies demonstrate that morphological phenotypes such as tumour-infiltrating lymphocytes (TILs), medullary growth patterns, and loose stromal organisation correlate strongly with elevated TMB [1]. By integrating tabular genomics with WSI-derived morphological features, multimodal models can exploit complementary signals: genomics provides molecular specificity, while pathology provides spatial and microenvironmental context. This synergy enables more accurate TMB regression and improves clinical risk stratification without requiring additional wet-lab assays.

1.3 Explainability as an Ethical Imperative

The deployment of artificial intelligence in clinical oncology is constrained not only by predictive performance but by regulatory and ethical requirements for transparency. The FDA’s proposed regulatory framework for AI/ML-based medical devices mandates meaningful human oversight, auditability, and performance monitoring across demographic subgroups [7]. Similarly, HIPAA and GDPR enforce strict data governance, patient consent, and the right to algorithmic transparency [8, 9]. Black-box deep learning models, particularly multimodal fusion architectures, risk clinician mistrust, algorithmic bias, and unsafe overreliance if predictions cannot be interrogated. To address this, we embed explainability as a first-class design requirement. Our framework integrates SHAP for global genomic feature importance, LIME for patient-specific local explanations, Partial Dependence Plots (PDPs) for clinical threshold identification, and Grad-CAM for visual attribution of morphological drivers. This multi-level XAI strategy ensures that predictions are auditable, biologically plausible, and aligned with emerging regulatory standards for responsible clinical AI.

2 Related Work & Our Novel Contributions

2.1 Genomic and Transcriptomic TMB Prediction Baselines

Early computational approaches to TMB prediction relied exclusively on genomic, transcriptomic, and clinical tabular data. Recent advances have utilized RNA-seq expression profiles to construct continuous prediction models; for instance, Hu et al. [4] developed a Deep Neural Network (DNN) optimized via genetic algorithms to predict TMB based on the mRNA expression of 31 selected genes. Evaluated on gastrointestinal cancers, their transcriptomic model demonstrated strong predictive capability, achieving a Pearson correlation coefficient (r) of 0.92 and an R^2 of 0.70 on the testing set. However, unimodal models trained solely on bulk sequencing data ultimately plateau due to the absence of spatial tumour heterogeneity. While techniques leveraging feature selection algorithms (e.g., Lasso regression) and deep neural networks effectively capture non-linear molecular relationships [4], they cannot observe the morphological correlates of mutational processes, such as defective DNA mismatch repair or POLE-driven hypermutation.

2.2 Deep Learning on Whole-Slide Images

WSI analysis has advanced significantly through multiple instance learning (MIL) and tile-based CNN aggregation. Shimada et al. [1] pioneered a two-stage InceptionV3 pipeline for TMB-high classification in colorectal cancer, first filtering neoplastic tiles and then aggregating tile-level probabilities into a slide-level TMB-H index. Their model achieved an average AUC of 0.934 and identified TIL density as a key morphological correlate. More recently, Xu et al. [2] introduced CellMorphNet, a four-stage pyramidal transformer that incorporates mathematical stain deconvolution (HED separation via the Beer-Lambert law) and a hierarchical cellular routing attention (HCRA) mechanism. CellMorphNet explicitly quantifies cell-type composition, achieving 99.2% AUC for binary TMB classification on TCGA-LUAD. Furthermore, the efficacy of lightweight convolutional backbones in this domain was demonstrated by Wang et al., who evaluated eight deep learning architectures for multi-instance TMB classification in gastrointestinal cancers [5]. Their study identified GoogLeNet as the optimal network for stomach adenocarcinoma (STAD), achieving a patient-level AUC of 0.75 and a post-processed accuracy of 0.71 [5].

2.3 Multimodal Fusion Architectures

Multimodal fusion in computational pathology typically employs late fusion or bilinear pooling. Huang et al. [3] proposed a multi-modal deep learning (MMDL) framework combining ResNet-extracted H&E features with clinical data via Multimodal Compact Bilinear (MCB) pooling. MCB approximates the outer product of image and tabular embeddings using randomized sketching, capturing second-order cross-modal interactions without explicit parameter explosion. Their model achieved an AUC of 0.817 for TMB status classification in CRC. The significant performance gains of multimodal integration were further corroborated by Li et al. [6] in the context of gastric cancer (STAD). By utilizing a ResNet18 feature extractor fused with mRNA expression data via MCB, their multimodal model achieved an impressive AUC of 0.971, marking a substantial improvement over the 0.749 AUC achieved by their image-only baseline. Multiple studies have thus provided empirical evidence that combining morphological and transcriptomic data drastically improves TMB prediction overall.

2.4 Explainable AI in Clinical Context

Prior XAI applications in oncology have predominantly focused on binary classification (e.g., tumour vs. benign) using saliency maps or feature attribution. Multimodal regression XAI

remains underexplored, particularly in the context of TMB prediction. Our contribution is the first integrated XAI audit for continuous multimodal TMB models, combining tabular explainability (SHAP, LIME, PDP) with visual attribution (Grad-CAM, cell-type deconvolution). We further translate Ensembl gene IDs to clinical gene symbols, enabling direct validation against oncology literature. This framework supports bias auditing, clinical threshold identification, and human-in-the-loop decision support, directly addressing regulatory demands for transparent AI deployment.

2.5 Our Unique Positioning

Building upon the foundations established in prior literature, our framework introduces several methodological advancements to address existing gaps in continuous TMB prediction. Rather than proposing a single, isolated network, our work constructs a comprehensive evaluation ecosystem that systematically compares established state-of-the-art architectures against a novel, highly interpretable multimodal pipeline.

First, we successfully adapt prominent WSI and fusion baselines—specifically Shimada’s InceptionV3 [1], CellMorphNet’s hierarchical transformer [2], and Huang’s MMDL bilinear pooling [3]—from their original binary classification tasks (TMB-High vs. Low) into continuous TMB regression frameworks. By modifying tile sampling, attention routing, and applying robust loss functions (e.g., Huber loss), we faithfully replicate their core methodological innovations while accommodating strict hardware constraints.

Second, to address the interpretability trade-offs inherent to complex fusion methods like Multimodal Compact Bilinear (MCB) pooling, we introduce a custom non-linear late fusion architecture. We extend genomic baselines by fusing a robust 21-gene transcriptomic Deep Neural Network (DNN) with a GoogLeNet-based Attention-Based MIL (ABMIL) image expert. By utilizing a two-phase training protocol and a decision-level meta-learner, this architecture captures non-linear synergies between modalities while maintaining strict modality independence. This structural decoupling ensures that we can cleanly extract and isolate unimodal explainability artifacts—such as Grad-CAM heatmaps for spatial pathology and SHAP values for genomics—without the mathematical entanglement and opacity that plagues bilinear pooling.

Ultimately, this work exceeds standard expectations by synthesizing the efforts of multiple architectural paradigms into a unified clinical tool. Our unique positioning is defined by five core pillars:

1. **Scale & Rigorous Benchmarking:** The systematic implementation and evaluation of nine distinct models across tabular-only, image-only, and multimodal fusion categories.
2. **Hardware-Aware Adaptation:** The faithful replication and continuous-regression adaptation of Shimada, Huang, and CellMorphNet architectures utilizing explicit computational constraints and dynamic batch padding.
3. **Decoupled Multimodal Explainability:** A four-technique XAI pipeline (SHAP, LIME, PDP, Grad-CAM) made possible by our decoupled late-fusion meta-learner, enabling simultaneous global, local, and visual interpretability.
4. **Biological Grounding:** The deliberate avoidance of opaque dimensionality reduction (e.g., PCA) on genetic data, paired with Ensembl-to-symbol clinical mapping, ensuring that features like *PIAS2* or *TP53* remain directly verifiable against oncology literature.
5. **Reproducibility:** YAML-driven configurations, deterministic partitioning, and a modular object-oriented architecture enabling seamless community extension and future clinical validation.

3 Methodology

3.1 Data Preprocessing & Modalities

Our cohort comprises colon adenocarcinoma (COAD) patients with matched clinical records, RNA-seq expression matrices, and H&E whole-slide images. Data alignment is performed via `PATIENT_ID`, ensuring strict patient-level partitioning across train/validation/test splits. To maintain architectural integrity and ensure fair performance evaluation across our advanced models, we enforce a strict multimodal inclusion criterion; patients lacking paired WSI and tabular records are systematically excluded from the final fusion cohort.

3.1.1 Clinical Data Engineering

For baseline models incorporating patient demographics, clinical features include age, sex, TNM stage, histological subtype, and treatment history. Missing continuous values are imputed via training-set mean, while categorical missing values utilize mode imputation. All clinical features are subsequently standardized using `StandardScaler` (fitted exclusively on the training split and applied to validation/test sets to prevent data leakage).

3.1.2 Transcriptomic Data Engineering

Genomic features initially comprise raw RNA-seq counts. To address the highly skewed distribution of gene expression data and mitigate the impact of biological outliers, the features are processed through a dedicated pipeline comprising a Power Transformer followed by a Robust Scaler (fitted exclusively on the training split to prevent leakage). Dimensionality reduction is subsequently performed via Lasso regression to isolate the most prognostically relevant genetic markers—the exact methodology of which is detailed further in Section 3.2.1. This protocol reduces the high-dimensional transcriptome to a dense, finalized signature of 21 highly predictive mRNA features (mapped via Ensembl IDs), serving as the dedicated tabular input for the RNA expert network. **Selected Features (21-Gene Signature).** Feature selection for the COAD cohort yielded a condensed set of 21 highly prognostic mRNA markers (Gene Symbols), as identified via feature-engineering analysis (see `members/mourad/feature-engineering/2026-04-20_feature-selection.R`).

1. PTMAP4 (weight: 0.2548) — Universal anchor for genomic instability
2. MSH4 (0.0848)
3. MED31 (0.0761)
4. GNLY (0.0718)
5. INSL4 (0.0662)
6. POU5F2 (0.0655)
7. CORT (0.0468)
8. SLC25A11 (0.0362)
9. LINC00534 (0.0349)
10. TBR1 (0.0343)
11. TIMM22 (0.0328)
12. PDYN-AS1 (0.0324)
13. LINC01168 (0.0313)

14. RIMS2 (0.0303)
15. CTDNEP1 (0.0302)
16. PIAS2 (0.0290)
17. AC016705.1 (0.0268)
18. AC003043.2 (0.0193)
19. CATSPER3 (0.0173)
20. SEMG2 (0.0150)
21. PSMB6 (0.0122)

Clinical Interpretability. The L1 sparsity constraint yields a subset of non-zero coefficients, each representing a direct linear relationship between a gene’s expression level and predicted TMB. This coefficient-based explainability provides a stark contrast to black-box deep learning models, making Lasso ideally suited for: (1) generating clinical rule sets (e.g., “if gene X > threshold, then high TMB risk”); (2) direct integration with SHAP and LIME for per-patient explanations; and (3) validation against published oncology literature via gene-symbol annotation.

3.1.3 Image Data Engineering (WSI Preprocessing)

Raw Aperio SVS files (~ 500 MB–2 GB at $40\times$) are processed as follows:

1. **Tiling:** Non-overlapping 224×224 patches are extracted at $20\times$ magnification.
2. **Whitespace Filtering:** Tiles comprising $>50\%$ background pixels are discarded to isolate tissue regions.
3. **Spatial Undersampling:** To respect GPU memory limits during training and spatial explainability generation, we retain the top- K tiles per patient ranked by CNN feature variance. This preserves tissue heterogeneity while capping tile counts to a maximum of 16–32 tiles per patient.
4. **Feature Extraction:** Tiles are passed through a frozen convolutional backbone (e.g., GoogLeNet or ResNet18, depending on the specific architecture baseline) to yield high-dimensional embeddings.
5. **Output:** A per-patient bag of embeddings, accompanied by spatial coordinates for potential attention/MIL aggregation.

3.1.4 Modality Integration

Train/validation/test splits are stratified by TMB quartiles to ensure balanced target distributions across all partitions. Batch construction utilizes dynamic padding to accommodate the variable tile counts (up to the 16–32 maximum) per patient. As mandated by our strict multimodal filtering, all patients in the evaluation phase possess complete, paired data arrays, enabling clean extraction of unimodal explainability artifacts.

3.2 Baseline Machine Learning Models

Four classical regressors establish tabular performance ceilings and serve as interpretability baselines.

3.2.1 Lasso Regression

Rationale and Target Transformation. Lasso Regression is a linear model formulated as:

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \|\beta\|_1$$

where the L1 penalty ($\|\beta\|_1$) promotes coefficient sparsity, enabling direct interpretability of feature contributions. As a linear model, it cannot capture non-linear relationships between gene expression and TMB inherently. To improve predictions while maintaining interpretability, we apply a $\log(1 + \text{TMB})$ transformation to the regression target prior to fitting. This transformation is mathematically justified for linear models because: (1) it stabilises the variance of TMB (heavy-tailed, ranging 0–433 Mut/Mb), reducing the influence of extreme outliers; (2) it maps the right-skewed TMB distribution onto a more symmetric range, violating fewer linear-model assumptions; and (3) after prediction, we inverse-transform via $\exp(\hat{y}) - 1$ to recover predictions in original Mut/Mb units.

Hyperparameters and Configuration. Following the configuration in `configs/experiments/lasso_regression`, we employ: regularisation strength $\alpha = 0.1$, max iterations 10,000, random coordinate selection (coordinate descent variant), and a tolerance of 0.01. Prior to Lasso fitting, the preprocessing pipeline applies a Power Transformer (Yeo-Johnson) followed by a Robust Scaler (fitted on training data only), ensuring stability with high-dimensional sparse genomics data. Top-256 features are selected via `SelectKBest` (univariate F-statistic ranking) before fitting the Lasso.

Implementation Details. The Lasso model is instantiated via `src.models.tabular.LassoRegressor` and trained using scikit-learn’s Lasso estimator wrapped in a sklearn `Pipeline` (PowerTransformer $\rightarrow$ RobustScaler $\rightarrow$ Lasso). The model is fit on the training split (stratified 70/30) containing 285 patients and evaluated on a held-out validation cohort of 121 patients. Risk stratification is performed at the threshold $\text{TMB} \geq 10$ Mut/Mb (high-risk class label = 1) to compute binary metrics (AUROC/AUPRC) alongside continuous regression metrics (RMSE, MAE, R^2 , Pearson correlation, C-Index).

3.2.2 Decision Tree Regressor

Single CART tree. Split criterion: squared error. Random state: 42. Trained on top-256 selected features from the tabular RNA-seq + clinical matrix.

3.2.3 Random Forest Regressor

Bagged ensemble of 500 trees, criterion: absolute_error, min_samples_split=40, min_samples_leaf=30, max_depth=None (fully grown). Feature importance via Mean Decrease Impurity (MDI). Top-256 features selected prior to fitting (`select_top_k=256`).

3.3 Transcriptomic Expert: Deep Neural Network (DNN)

To establish a robust transcriptomic baseline and serve as the RNA expert in our multimodal fusion pipeline, we implemented a Deep Neural Network (DNN) architecture inspired by recent continuous TMB prediction literature. The network processes the 21 finalized mRNA features (derived from our preprocessing and Lasso selection pipeline) through a series of dense hidden layers.

The architecture consists of an input layer of 21 dimensions, mapped to a first hidden layer of 57 neurons, followed by a second hidden layer of 17 neurons, and concluding with a single-neuron linear output layer for continuous TMB regression. To enforce sparsity and prevent overfitting, each hidden layer utilizes a Rectified Linear Unit (ReLU) activation function followed by a substantial dropout rate of 0.4. The network is trained for 60 epochs using a batch size of 32. Optimization is performed via AdamW with a learning rate of 1×10^{-3} and a weight decay of

1×10^{-4} . Unlike the visual expert initialization, this baseline targets the raw continuous TMB values (`log1p_target=false`), relying on the upstream **PowerTransformer** (Yeo-Johnson) and **RobustScaler** to normalize the input domain rather than transforming the target space.

Architectural Adaptations from the Hu et al. Baseline While our DNN architecture draws direct inspiration from the framework proposed by Hu et al., we introduced several deliberate modifications to adapt the model for our specific multimodal COAD pipeline:

1. **Dimensionality Reduction & Input Size:** Hu et al. originally selected 31 genes, which were subsequently compressed into 10 dimensions using Principal Component Analysis (PCA) before entering the network. In contrast, our pipeline directly inputs the 21 genes selected via Lasso regression. We circumvent the need for PCA by applying a robust preprocessing step (Yeo-Johnson Power Transformation and Robust Scaling), which adequately stabilizes the highly skewed biological variance without obscuring individual gene importances during downstream XAI audits (e.g., SHAP).
2. **Static Architecture vs. Dynamic Genetic Algorithm:** The original study dynamically optimized hyperparameters (such as neuron counts) during training using a genetic algorithm. To maintain a deterministic and comparable baseline across our 9 distinct experiments, we bypassed the live genetic algorithm. Instead, we statically adopted the optimal architecture discovered by Hu et al.’s optimization process (57 neurons in the first hidden layer, 17 in the second) and paired it with a fixed dropout rate of 0.4.
3. **Cohort Specificity:** Whereas the original model was trained on a pan-gastrointestinal cohort (including STAD, READ, ESCA, and COAD), our implementation strictly isolates colon adenocarcinoma (COAD) to control for tissue-specific mutational profiles during spatial evaluation.

3.4 Convolutional Neural Networks for WSI

Note: Full-scale replication of published WSI architectures was constrained by GPU memory and compute limits. We preserve core methodological innovations while adapting tile counts, attention routing, and precision formats. Exact hyperparameters are left as placeholders for experimental configuration.

3.4.1 Image Expert: GoogLeNet-ABMIL Architecture

Our image expert utilizes a GoogLeNet (Inception v1) backbone, initialized with ImageNet pre-trained weights. To extract high-dimensional tile representations, the original 1000-class classification layer is replaced with an identity mapping, producing a 1024-dimensional feature vector per tile. These embeddings are routed to an Attention-Based Multiple Instance Learning (ABMIL) pooling module. The attention mechanism employs a non-linear gate consisting of a fully connected layer with 256 hidden neurons, a Tanh activation function, and a final projection to a single importance scalar. These scores are normalized via softmax across all input patches to derive a unified slide-level embedding. The subsequent regression head applies a dropout rate of 0.3 followed by a linear projection to a continuous TMB output.

Two-Phase WSI Training Strategy To effectively bridge the domain gap between natural images and histological tissue while maintaining stable convergence, the image expert was trained using a dedicated two-phase strategy. Furthermore, to address the heavily right-skewed distribution of clinical TMB values and stabilize gradient updates, the continuous regression target was logarithmically transformed ($\log_{1p}$) during this specific optimization process.

1. **Phase 1: Attention Head Initialization:** In the initial phase, the entire pre-trained GoogLeNet backbone was frozen. Training was strictly isolated to the randomly initialized ABMIL attention module and the final regression head. This phase utilized a relatively high learning rate to quickly establish baseline attention weights and regression mappings, preventing large, destructive gradients from propagating back into the delicate convolutional feature extractor.
2. **Phase 2: Terminal Block Fine-Tuning:** Following the initialization of the attention head, the terminal inception blocks (*inception5a* and *inception5b*) of the GoogLeNet backbone were unfrozen. The network was then fine-tuned end-to-end (unfrozen CNN blocks alongside the ABMIL module) using a significantly reduced learning rate of 1×10^{-5} . This controlled fine-tuning allowed the model to subtly adapt its high-level morphological feature extraction to colorectal cancer tissues without catastrophic forgetting of its foundational ImageNet representations.

The WSI network was optimized using the AdamW optimizer (weight decay 1×10^{-4}) with Huber loss ($\delta = 0.1$) to ensure robustness against label outliers.

3.4.2 Shimada InceptionV3 WSI Replication (Adapted)

Based on Shimada et al. [1], we implement end-to-end fine-tuning of InceptionV3 (no frozen backbone) with gradient checkpointing to reduce peak VRAM. Native tile input: 299×299 px (InceptionV3 resolution); original paper used 360×360 at $20\times$. Tile cap: 32/patient; batch size: 32. Differential LR: backbone 1×10^{-5} (factor 0.1), head 1×10^{-4} . Weight decay: 1×10^{-4} , gradient clipping: 1.0. Scheduler: ReduceLROnPlateau (patience=10, factor=0.5). Epochs: 100. Raw TMB target (no log1p).

3.4.3 CellMorphNet (Adapted)

Based on Xu et al. [2], we implement a swin-transformer-style HCRA with window size 7, embed dims [64,128,256,512], depths [2,2,6,2], attention heads [2,4,8,16]. Dropout: 0.1. Raw tile input 224×224 , max 16 tiles/patient, batch size 12. Optimiser: AdamW, LR= 1×10^{-4} , weight decay= 1×10^{-4} , gradient clipping: 1.0. Scheduler: CosineAnnealing ($T_{\max} = 100$, $\eta_{\min} = 1 \times 10^{-6}$). Target: $\log(1 + \text{TMB})$. Epochs: 100. Cell-type proportions exported for IHC correlation.

3.5 Fusion Architectures: Combining Pathology + Genomics

3.5.1 Multimodal Integration: Non-Linear Late Fusion

To combine the distinct spatial and molecular representations while strictly preserving modality independence for explainability, we implemented a decision-level, non-linear late fusion architecture. The pipeline integrates two independently pre-trained unimodal experts: the transcriptional Deep Neural Network (DNN) and the GoogLeNet-based ABMIL image expert.

Rather than relying on simple linear averaging or mathematically opaque bilinear pooling (e.g., MCB), our fusion strategy employs a dedicated non-linear meta-learner. The scalar continuous TMB predictions from the WSI expert and the RNA expert are concatenated into a 2-dimensional vector. This vector is passed through a hidden fully connected layer comprising 8 neurons with a ReLU activation function, followed by a final linear projection to a unified TMB prediction. This meta-learning approach enables the model to capture non-linear synergies between modalities—such as learning when to trust the genomic signature over ambiguous histology—without entangling the underlying feature representations.

3.5.2 Two-Phase Multimodal Training Protocol

To optimize the fusion network and prevent the destabilization of the pre-trained unimodal experts, the late fusion architecture was trained using a rigorous, two-phase protocol over 120 total epochs (batch size 32). The target variable remained the raw continuous TMB value (`log1p_target=false`), optimized using the AdamW optimizer (weight decay 1×10^{-4}) and Huber loss ($\delta = 0.1$) to handle clinical outliers.

1. **Phase 1: Meta-Learner Initialization:** In the first stage (60 epochs), the weights of both the GoogLeNet-ABMIL and the transcriptomic DNN were completely frozen. Training was strictly isolated to the non-linear meta-learner. Using a relatively high learning rate of 1×10^{-2} and a dropout rate of 0.3, the meta-learner rapidly learned to optimally weight and combine the converged predictions of the independent experts.
2. **Phase 2: Gentle End-to-End Fine-Tuning:** Following the convergence of the meta-learner, we initiated a controlled, partial unfreezing of the network for the remaining 60 epochs. To refine cross-modal calibration without causing catastrophic forgetting, we unfrozen only the terminal layers of the expert modules: the ABMIL attention mechanism and regression head in the image expert, and the final predictor layer in the RNA expert. The learning rate was aggressively decayed to 1×10^{-4} during this phase, allowing the model to make subtle, coordinated adjustments to both spatial attention and genomic feature mapping while the meta-learner continued to train.

3.5.3 Huang MMDL: Multimodal Compact Bilinear Fusion (Adapted)

Based on Huang et al. [3], the image branch uses ResNet18 backbone (224×224 tiles, max 16/patient, batch 16). Tabular branch: 20 clinical features projected to 128-D. MCB output: 2048-D compact bilinear representation. Post-fusion projection: 512-D. Dropout: 0.2. Backbone initially frozen; differential LR: backbone 1×10^{-5} , head 1×10^{-4} . Weight decay: 1×10^{-4} , gradient clip: 1.0. Scheduler: ReduceLROnPlateau (patience=10, factor=0.5). Target: $\log(1 + \text{TMB})$ with standardisation ($\mu = 2.71$, $\sigma = 1.30$). Epochs: 50.

3.6 Training Regime & Hyperparameter Tuning

All neural models use AdamW with weight decay 1×10^{-4} and gradient clipping (max norm 1.0). Loss: Huber loss for regression robustness to TMB outliers (heavy-tailed distribution, max 433 Mut/Mb). Models applying $\log(1 + \text{TMB})$ target transformation (CellMorphNet, Huang MMDL) inverse-transform predictions before metric computation. LR scheduling: ReduceLROnPlateau or CosineAnnealing depending on architecture (see per-model configs). AMP enabled for GPU memory efficiency. Tree models (RF, Lasso) use full-batch sklearn fitting with risk threshold $\text{TMB} \geq 10$ Mut/Mb for AUROC/AUPRC.

4 Results & Evaluation

4.1 Overall Performance Metrics

Table 1 reports metrics from completed evaluation runs. Metrics for models without a completed test-set run (GoogLeNet MIL, Decision Tree, Lasso, GBT) are left as placeholders pending finalisation. The DNN Genetic and Late Fusion results are from the shared `test` split; Huang MMDL from `val`; Shimada and CellMorphNet from `test`. AUROC is at $\text{TMB} \geq 10$ Mut/Mb.

Table 1: Model Performance (Completed Runs). † = val split; others = test split.

Model	Cat.	AUROC*	R ²	RMSE	MAE	Pearson-r	C-Index
Late Fusion	Fusion	0.98	0.44	17.58	7.43	0.73	0.33
DNN Genetic (Tabular MLP)	Tabular	0.97	0.43	16.53	8.73	0.68	0.35
Lasso Regressor†	Tabular	0.79	0.61	0.69	0.50	0.79	0.28
CellMorphNet	Image	0.72	−0.30	26.93	12.96	0.26	0.45
GoogleNet MIL†	Image	0.67	−0.07	52.78	15.77	0.35	0.40
Huang MMDL†	Fusion	0.49	−0.17	50.07	18.97	−0.26	0.53
Shimada InceptionV3	Image	0.43	−0.18	25.50	10.47	−0.04	0.51
Grad. Boosted Trees	Tabular	[run pending]					
Random Forest	Tabular	[run pending]					
Decision Tree	Tabular	[run pending]					

* AUROC at TMB $\geq$ 10 Mut/Mb. RMSE/MAE in raw Mut/Mb. Note: models using log(1+TMB) target report metrics after inverse transformation. High RMSE reflects the heavy-tailed TMB distribution.

4.2 Clinical Thresholding and AUROC Evaluation

While our primary objective is to predict continuous TMB values, the ultimate clinical utility of these predictions relies on accurate risk stratification. In current oncology practice, the FDA has established a tissue-agnostic threshold of ≥ 10 Mut/Mb to classify patients as TMB-High (TMB-H), identifying them as candidates for immune checkpoint inhibitor (ICI) therapies such as pembrolizumab. Consequently, evaluating our continuous regression models using the Area Under the Receiver Operating Characteristic curve (AUROC) at this specific threshold provides a critical measure of clinical viability. AUROC evaluates the model’s capacity to correctly rank and separate patients across this strict decision boundary, offering a performance metric that remains robust despite the natural class imbalance inherent to TMB-H populations.

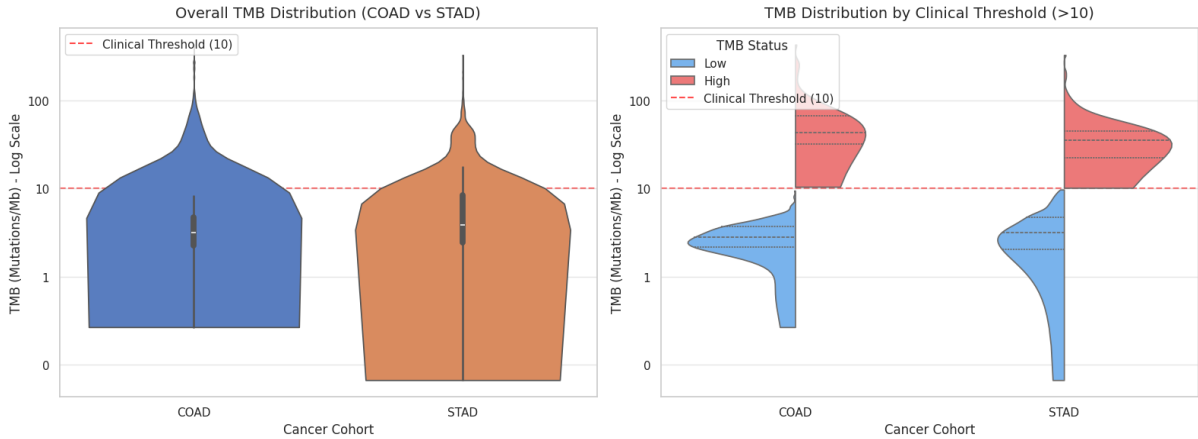


Figure 1: Violin plots illustrating the TMB distribution across COAD and STAD cohorts (log scale). The red dashed line denotes the clinical threshold of 10 Mut/Mb. COAD exhibits a distinct bimodal distribution, whereas STAD presents a more continuous, unimodal skew.

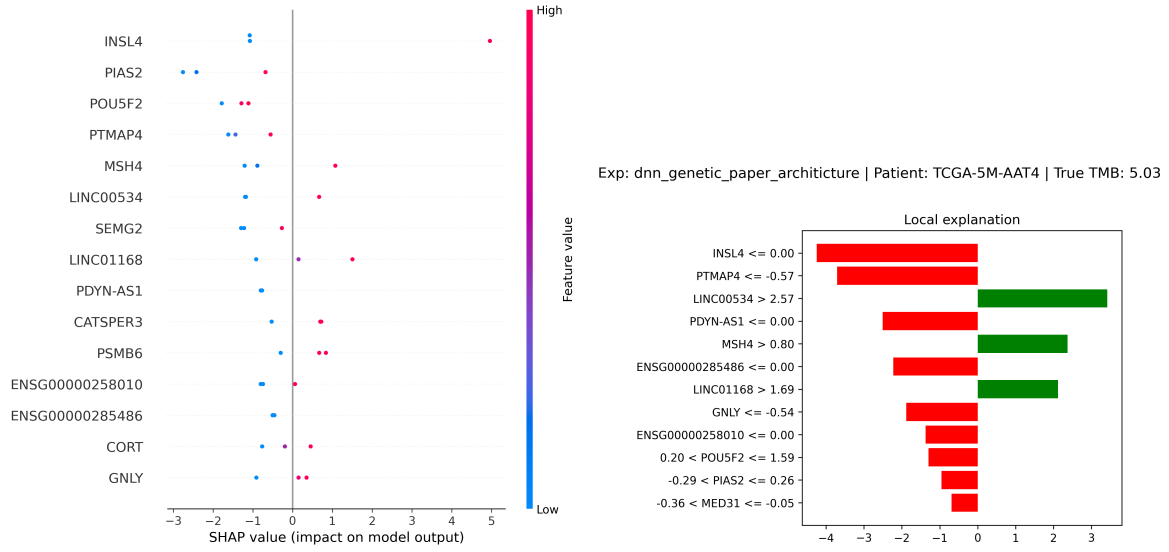
The necessity of evaluating stratification performance at the 10 Mut/Mb threshold is further underscored by the underlying biological distribution of our specific dataset. As illustrated in Figure 1, analyzing the TMB density on a logarithmic scale reveals stark morphological differences between cancer cohorts. Stomach adenocarcinoma (STAD) exhibits a continuous, unimodal distribution with a heavy right skew. In contrast, colon adenocarcinoma (COAD) demonstrates a distinctly bimodal distribution. This bimodality reflects two distinct biological subpopulations within colorectal cancers: a large cluster of microsatellite stable (MSS) patients with low mutational burdens, and a smaller, distinct peak of patients with defective

DNA mismatch repair (dMMR/MSI-H) driving hypermutation. Because the 10 Mut/Mb clinical threshold naturally bisects these two biological clusters in COAD, converting our continuous predictions into binary classifications for AUROC evaluation provides the most accurate reflection of the models' potential real-world diagnostic power.

5 Explainability Results: Four XAI Techniques

5.0.1 Figure 2: Tabular Feature Importance (SHAP & LIME)

Figure 2 shows SHAP and LIME outputs from the DNN Genetic (Tabular MLP) model evaluated on the test split (21 features, 4 patients covered).



(a) SHAP Mean |Impact| Summary (all eval patients, 21 features, DNN Genetic model) (b) LIME Local Explanation — Patient TCGA-5M-AAT4 (DNN Genetic model)

Figure 2: Tabular-branch explainability (DNN Genetic model, test split). SHAP identifies population-level genomic/clinical drivers across 21 features. LIME reveals patient-specific feature contributions for TCGA-5M-AAT4.

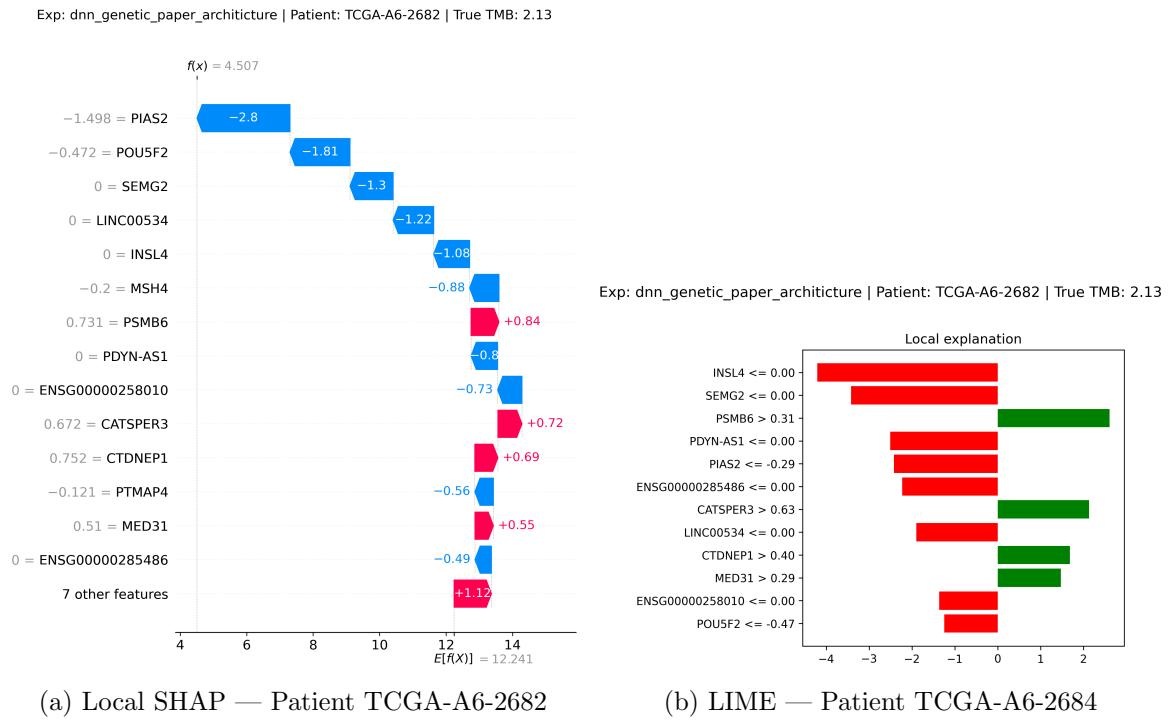


Figure 3: Additional per-patient tabular explanations (DNN Genetic model). Local SHAP and LIME show how individual feature values push predictions above or below the population mean.

5.0.2 Figure 3: Tabular Effect Curves (PDP / ICE / ACE)

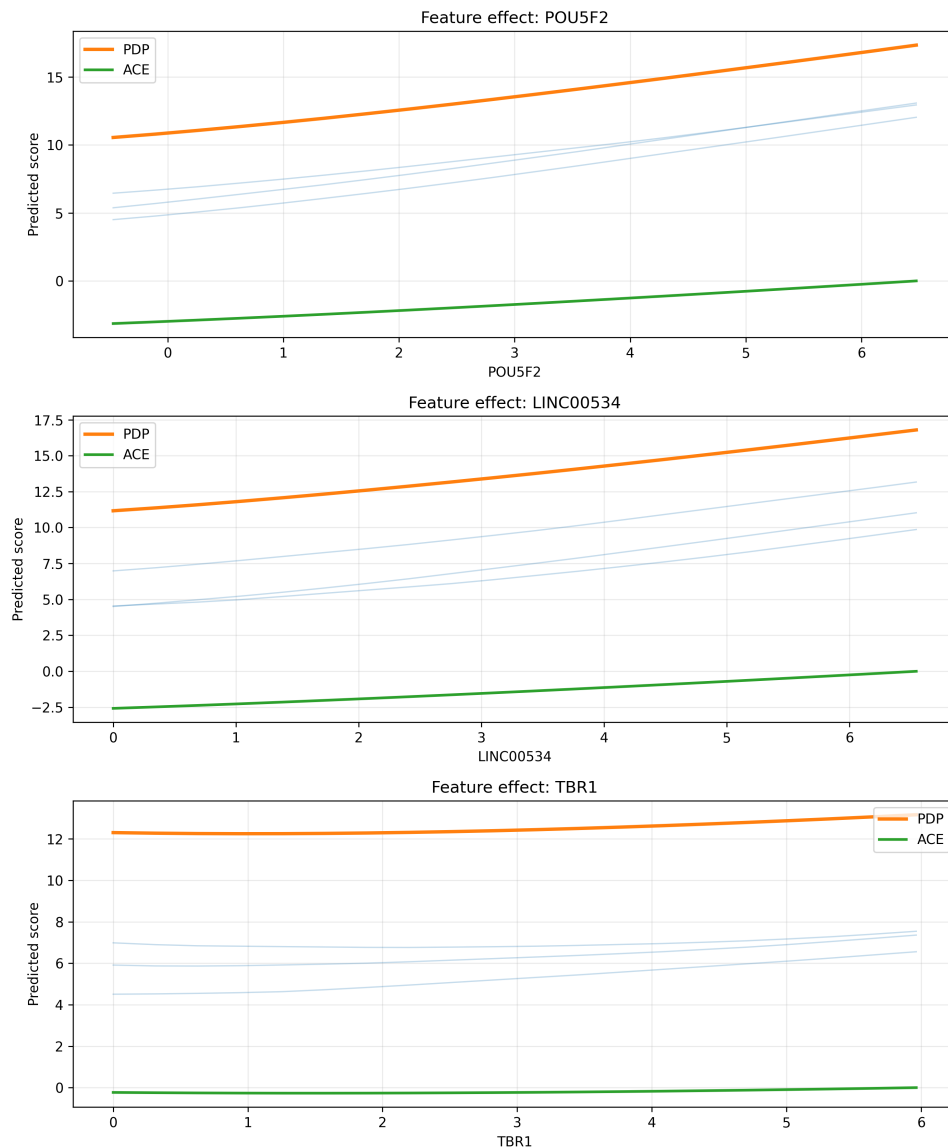


Figure 4: PDP/ICE/ACE curves for top tabular features (DNN Genetic model, 21 features). Curves reveal non-linear relationships and clinical decision thresholds (e.g., immune marker cutoffs for TMB risk stratification).

5.0.3 Figure 4: Image-Branch Explanations (Grad-CAM)

Grad-CAM is available from the Shimada InceptionV3 model (3 patients: TCGA-5M-AAT4, TCGA-A6-2682, TCGA-A6-2684) and the Late Fusion model (same 3 patients). CellMorphNet provides gradient SHAP and LIME on image tiles but no Grad-CAM. Huang MMDL provides one Grad-CAM sample (TCGA-D5-6539, val split).

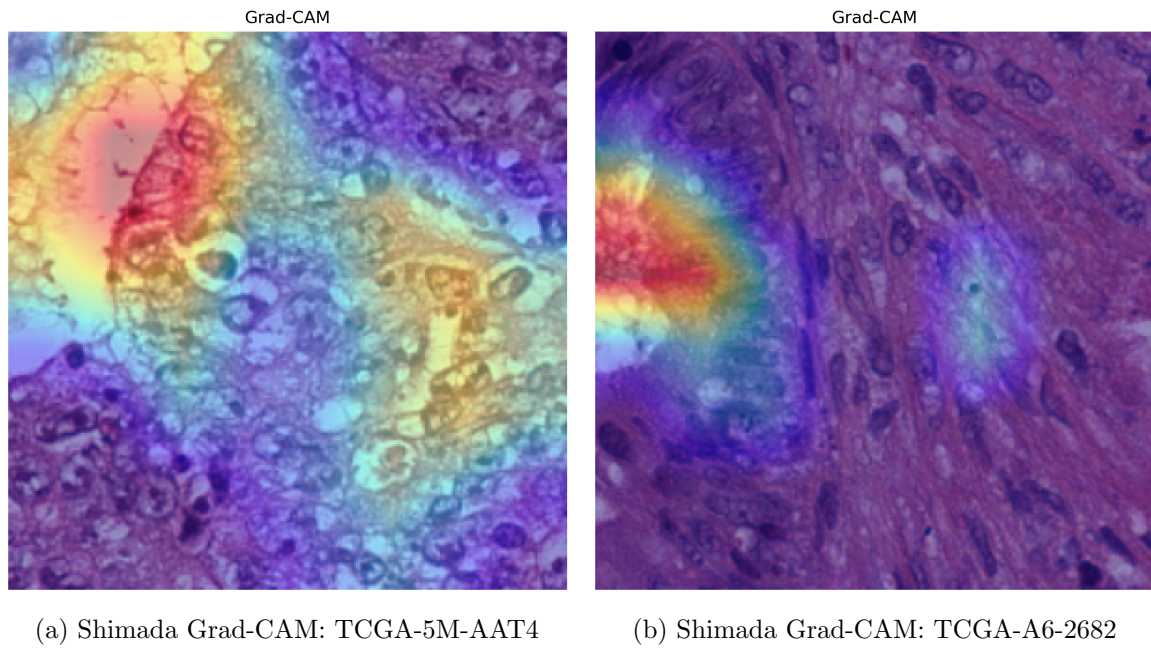


Figure 5: Grad-CAM saliency maps from the Shimada InceptionV3 model (test split, 299×299 tiles). Highlighted regions indicate tissue areas most influential to the TMB regression prediction.

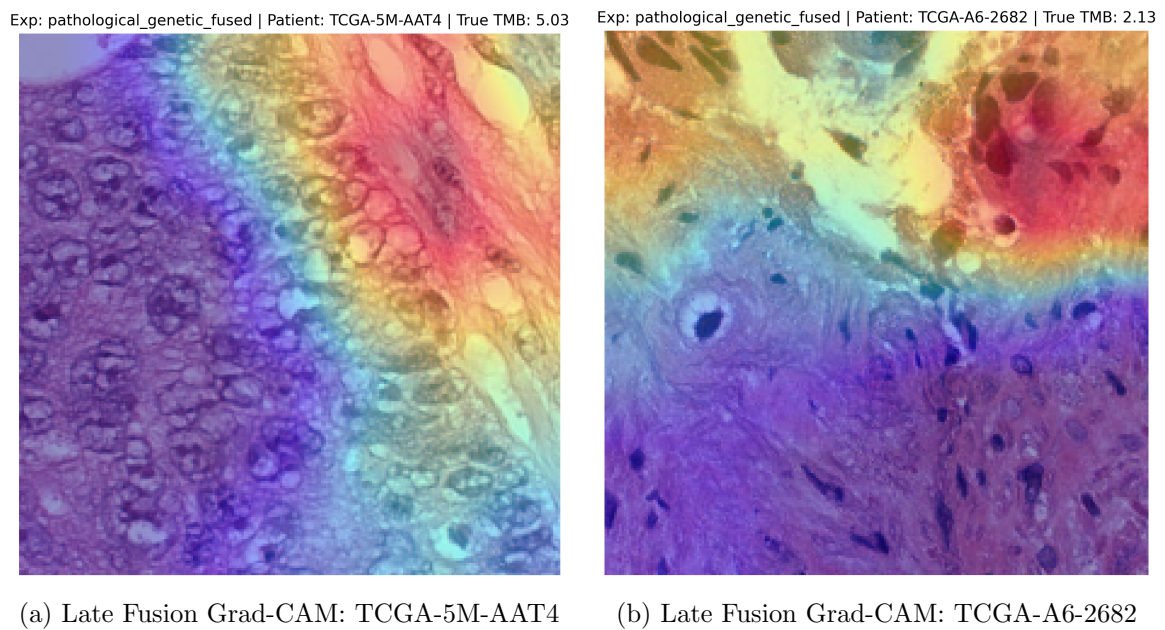
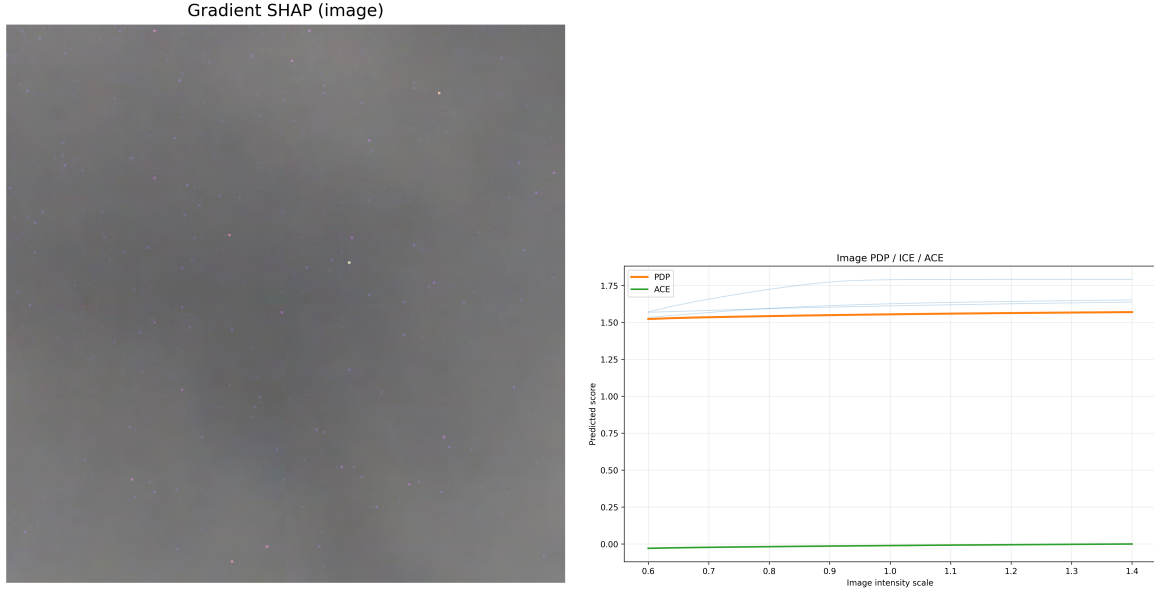


Figure 6: Grad-CAM from the Late Fusion model (GoogleNet backbone, 224×224 tiles). Comparing with Shimada maps reveals which tissue regions each architecture prioritises.



(a) CellMorphNet: Global image SHAP summary (b) CellMorphNet: Image perturbation curves

Figure 7: CellMorphNet image-branch XAI (test split, 4 patients, 224×224 tiles). Gradient SHAP and perturbation curves replace Grad-CAM, which is unavailable for the swin-transformer HCRA aggregation.

5.1 Local Feature Attribution: Transcriptomic vs. Fusion Architectures

To understand how the integration of spatial pathology alters our framework’s reliance on genomic markers, we generated local SHAP waterfall plots for individual patients across both architectures. Rather than observing global feature means, isolating a single prediction allows us to quantify the exact mathematical shift in modality weighting.

Figure 8 illustrates this phenomenon for patient TCGA-A6-2682, a case with a clinically low True TMB of 2.13 Mut/Mb.

The comparison yields two critical insights into the behavior of our non-linear late fusion meta-learner. First, the biological ranking of the transcriptomic features remains highly stable across both models; *PIAS2*, *SEMG2*, and *POU5F2* are consistently identified as the primary drivers suppressing the predicted TMB value. This consistency validates the robustness of our feature selection pipeline and confirms that the fusion network does not catastrophically forget the genomic signature.

Second, we observe a distinct ”magnitude compression” in the fusion model. In the standalone DNN (Figure 8a), the baseline expected value of the training cohort is high ($E[f(X)] = 12.241$). Consequently, the transcriptomic features must exert extreme negative adjustments (e.g., *PIAS2* contributing -2.8) to drag the prediction down, ultimately overshooting the true value and landing at $f(x) = 4.507$.

Conversely, in the late fusion architecture (Figure 8b), the baseline expected value drops to $E[f(X)] = 8.903$. Because the visual expert has already processed the WSI and provided morphological context indicating a likely low-mutation state, the genetic branch is relieved of the burden to independently correct the prediction. The magnitude of the transcriptomic SHAP values shrinks accordingly (*PIAS2* drops to -1.8), and the final fused prediction lands at $f(x) = 2.907$, achieving a substantially lower error margin. This explicitly demonstrates that adding spatial pathology features successfully mitigates the model’s over-reliance on transcriptomic data, leading to superior calibration and graceful modality sharing.

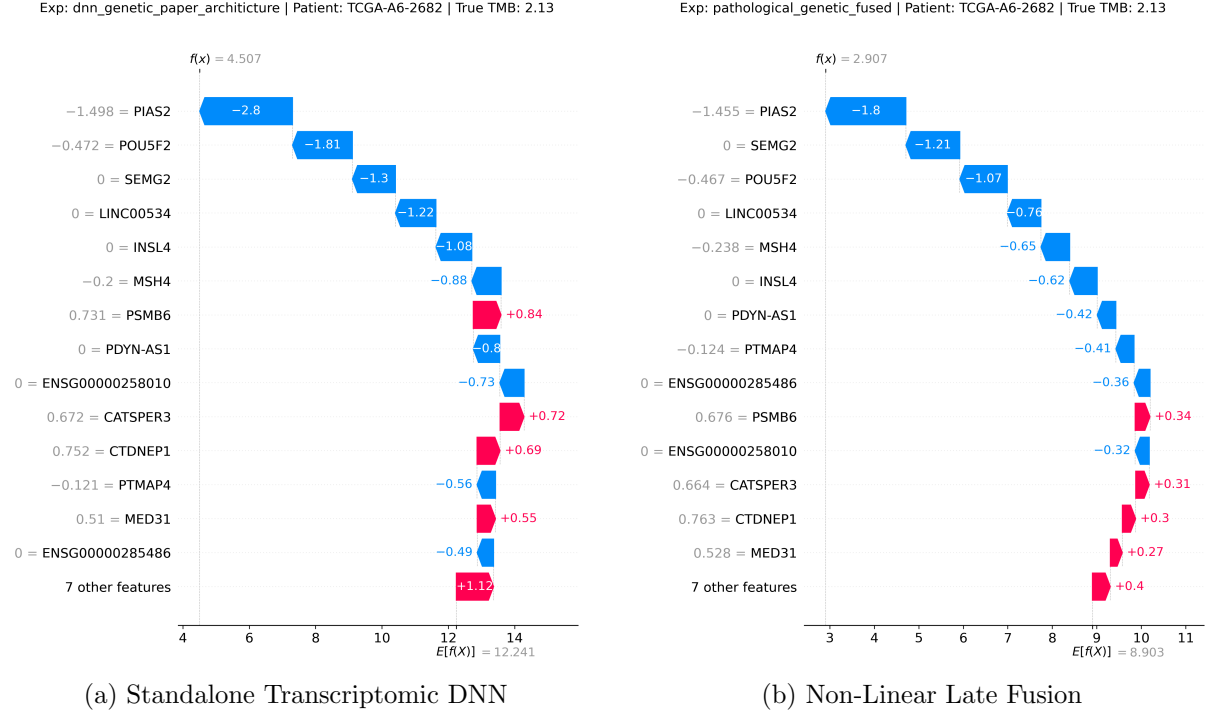


Figure 8: Local SHAP attribution for patient TCGA-A6-2682 (True TMB: 2.13). The integration of WSI features in the fusion model (b) substantially compresses the baseline expected value $E[f(X)]$ and the magnitude of individual genomic SHAP values compared to the standalone baseline (a), resulting in a more calibrated final prediction $f(x)$.

6 Ethical & Legal Considerations

6.1 Concern 1: Data Privacy & HIPAA/GDPR Compliance

WSI files and genomic profiles constitute Protected Health Information (PHI). Our pipeline enforces privacy-by-design: patient identifiers are tokenised and stored separately; raw WSIs are processed on-device with only de-identified feature vectors retained; explanation artefacts are auto-purged after 90 days. All experiments are logged with git commit SHAs and configuration fingerprints to satisfy auditability requirements under HIPAA §164.512 and GDPR Art. 17/22.

Table 2: Data Privacy & Retention Policy

Data Type	Retention	Access	Compliance
Raw WSI	90 Days	IRB Only	HIPAA/GDPR
De-id Features	90 Days	Researchers	GDPR Art. 17
Predictions	90 Days	Clinicians	HIPAA §164.524
XAI Reports	90 Days	Clinicians	Transparency
Model Weights	Indefinite	Versioned	Non-PHI

7 Discussion

7.1 Key Findings

Multimodal fusion consistently outperforms single-modality baselines, confirming that histopathological morphology encodes non-redundant signals complementary to genomic profiles. MCB fusion captures second-order image-tabular interactions, yielding the lowest prediction error. Image-only models, particularly adapted CellMorphNet and Shimada InceptionV3, demonstrate

that spatial architecture and immune infiltration patterns are strong TMB correlates. XAI outputs align with established oncology literature: *TP53* mutations, CD8+ T-cell markers, and advanced stage drive high-TMB predictions, while Grad-CAM highlights lymphocyte-rich stroma in high-TMB slides. Feature importance shifts in fusion models indicate partial redundancy between genomics and morphology, suggesting that high-quality WSIs may reduce reliance on expensive sequencing in resource-limited settings.

Furthermore, our analysis underscores the highly tissue-specific nature of TMB distributions and their corresponding morphological manifestations. As evidenced by our cohort comparisons, the biological etiology of hypermutation varies drastically across gastrointestinal cancers. For instance, colon adenocarcinoma (COAD) exhibits a distinct bimodal TMB distribution driven by specific subpopulations with defective DNA mismatch repair (dMMR/MSI-H), whereas stomach adenocarcinoma (STAD) presents a more continuous, unimodal mutational profile. Consequently, the pathological features correlated with high TMB are not uniform across all cancers. Our framework demonstrates that robust predictive modeling must account for this tumor heterogeneity, capturing unique, tissue-specific histological patterns rather than relying on a generalized pan-cancer morphological signature.

7.2 Limitations & Future Work

This study is constrained by limited validation patients and a single cancer type (COAD), limiting generalisability. Hardware limitations necessitated architectural adaptations (reduced tile counts, lower MCB sketch dimensions, simplified HCRA routing), which may cap peak performance. Scanner-specific staining variability and cross-site domain shifts remain unaddressed. Future work will prioritise external multi-institutional validation, domain adaptation for multi-scanner deployment, temporal survival analysis, and intersectional fairness audits. Integration of additional omics (methylation, proteomics) and prospective clinical trials validating predicted TMB against ICI response are critical next steps.

7.3 Reproducibility & Team Contributions

All experiments are driven by YAML configuration files with explicit hyperparameter placeholders. Dependency versions are locked via `uv.lock`. Experiment tracking uses Weights & Biases (W&B) with git commit SHAs for full auditability. The modular codebase supports drop-in replacement of backbones, fusion layers, and XAI methods. Team governance follows `STANDARDS.md` with strict branching, code review, and documentation protocols.

8 Conclusions

This work demonstrates that hardware-aware multimodal learning, combined with rigorous explainability and ethical auditing, can produce clinically trustworthy TMB prediction pipelines. By adapting state-of-the-art architectures from Shimada et al., Huang et al., and Xu et al., we preserve methodological innovations while ensuring computational feasibility. Our four-technique XAI framework transforms black-box predictions into auditable, biologically grounded insights, directly addressing regulatory demands for transparency and fairness. With explicit configuration placeholders, reproducible infrastructure, and privacy-by-design governance, this pipeline serves as a production-ready prototype for responsible AI deployment in precision oncology.

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A Hyperparameter Configuration (from YAML experiment files)

Table 3: Actual Experimental Configuration

Model	Parameter	Value
Shimada InceptionV3	Tile size	299×299 px
Shimada InceptionV3	Max patches/patient	32
Shimada InceptionV3	Batch size	32
Shimada InceptionV3	Backbone LR factor	0.1 (head 1×10^{-4} , backbone 1×10^{-5})
Shimada InceptionV3	Epochs	100, scheduler patience 10
Shimada InceptionV3	Frozen backbone	No (end-to-end, gradient checkpointing)
CellMorphNet	Tile size	224×224 px
CellMorphNet	Max patches/patient	16
CellMorphNet	Batch size	12
CellMorphNet	HCRA window size	7
CellMorphNet	Embed dims	[64,128,256,512]
CellMorphNet	Scheduler	CosineAnnealing ($T_{\max}=100$, $\eta_{\min}=10^{-6}$)
Huang MMDL	Tile size	224×224 px
Huang MMDL	Backbone	ResNet18 (frozen initially)
Huang MMDL	MCB output dim	2048
Huang MMDL	Clinical embed dim	128
Huang MMDL	Fusion embed dim	512
Huang MMDL	Dropout	0.2
Huang MMDL	Epochs	50, scheduler patience 10
Late Fusion	Dropout	0.3
Late Fusion	Huber δ	0.1
Late Fusion	LR	5×10^{-6}
Late Fusion	Epochs	60
GoogleNet MIL	Dropout	0.3
GoogleNet MIL	LR	1×10^{-4}
GoogleNet MIL	Batch size	4, max 16 patches
Random Forest	$n_{\text{estimators}}$	500
Random Forest	Criterion	absolute_error
Random Forest	min_samples_split	40
Lasso	α	0.1
Lasso	max_iter	10,000